A correlated prior on galaxy bias across thin redshift bins

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1 Introduction: why an intermediate bias model

A Fisher forecast of galaxy clustering in thin redshift bins must decide how to parametrize the galaxy bias of each bin. The bins belong to a few broad groups: the DESI tracer samples and their chunks, such as BGS, LRG1, LRG2, LRG3, ELG1 and ELG2. Within one group the galaxies are selected the same way, so their biases are expected to be similar but not identical.

The default model ties all bins of a group to a single amplitude. If $D(z)$ is the linear growth factor, normalized to $D(0) = 1$, the tied model sets

$$b_i \;=\; \frac{B_g}{D(z_i)}, \tag{1}$$

where b_i is the linear bias of bin i, z_i is the bin's redshift, and B_g is one free amplitude per group g. The $1/D$ evolution keeps the clustering amplitude $b(z)\,D(z)$ of the group constant: galaxies that stopped merging and follow the flow of matter ("passive evolution") behave this way.

The opposite extreme frees one bias per bin, with flat priors and no connection between neighbors ("independent"). The dark-energy figure of merit of the forecast depends on which extreme is chosen, and the question is how much of it rests on the tying itself. To answer that, this note builds the intermediate model: bins may deviate from the group's trend, but neighboring bins must deviate *together*.

The model multiplies the tied bias by a fractional deviation δ_i,

$$b_i \;=\; \frac{B_g}{D(z_i)}\,\big(1+\delta_i\big), \tag{2}$$

with B_g still free (flat prior). The deviations receive a Gaussian prior that is correlated inside a group,

$$\big\langle \delta_i\,\delta_j \big\rangle \;=\; \sigma_b^2\,\exp\!\left(-\,\frac{|z_i - z_j|}{L}\right), \tag{3}$$

and zero across groups. Here σ_b is the prior standard deviation of a single bin's fractional deviation, and L is the correlation length in redshift: bins much closer than L deviate almost as one, bins much farther apart deviate independently.

In the Fisher matrix the convenient parameters are the per-bin amplitudes

$$\beta_i \;\equiv\; B_g\,\big(1+\delta_i\big), \qquad \text{fiducial } \beta_i = B_g, \tag{4}$$

so that $b_i = \beta_i / D(z_i)$. Since $\beta_i - B_g = B_g\,\delta_i$, the prior covariance of the β_i within one group is

$$\mathsf{C} \;=\; s^2\,\mathsf{R}, \qquad s \;=\; \sigma_b\,B_g, \qquad \mathsf{R}_{ij} \;=\; \exp\!\left(-\,\frac{|z_i - z_j|}{L}\right). \tag{5}$$

The spread s carries the units of a bias amplitude; the correlation matrix R is dimensionless, with ones on the diagonal.

One subtlety drives most of this note. A plain Gaussian prior with covariance C would also penalize moving *all* β_i of a group up or down together, but that uniform shift is exactly the free amplitude B_g, which the tied model never constrains. Adding the plain prior would therefore make the comparison against the tied model unfair. The repair is to remove the uniform direction from the prior by a projection, and Sections 2–5 build that projection from the ground up.

2 Two bins first

Everything essential already happens with two bins. Call their amplitudes β_1 and β_2, with common fiducial value B, prior spread s per bin, and correlation coefficient r between the two deviations. The prior covariance and its inverse are

$$\mathsf{C} \;=\; s^2 \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}. \tag{6}$$

(The inverse is checked by multiplying: the off-diagonal of $\mathsf{C}\,\mathsf{C}^{-1}$ is $r - r = 0$ and the diagonal is $(1-r^2)/(1-r^2) = 1$.)

The correlation coefficient follows from the covariance kernel of Eq. (3). By definition,

$$r \;=\; \frac{\langle \delta_1 \delta_2 \rangle}{\sqrt{\langle \delta_1^2 \rangle\, \langle \delta_2^2 \rangle}} \;=\; \frac{\sigma_b^2\, e^{-\Delta z/L}}{\sigma_b^2} \;=\; e^{-\Delta z/L}, \tag{7}$$

with $\Delta z = |z_1 - z_2|$ the bin separation. The amplitude σ_b cancels: the correlation is set by the separation and L alone.

Now examine the prior penalty $\chi^2_{\rm prior} = \Delta\beta^{\mathsf{T}}\, \mathsf{C}^{-1} \Delta\beta$, where $\Delta\beta = (\beta_1 - B,\ \beta_2 - B)^{\mathsf{T}}$ is the displacement from the fiducial. Two kinds of move matter.

A uniform shift. Take $\Delta\beta = a\,(1,1)^{\mathsf{T}}$: both bins move up by the same amount a. Working out the quadratic form with Eq. (6),

$$\Delta\beta^{\mathsf{T}} \mathsf{C}^{-1} \Delta\beta \;=\; \frac{a^2}{s^2(1-r^2)} \left[\, (1-r) + (1-r) \,\right] \;=\; \frac{2a^2}{s^2\,(1+r)} \;\neq\; 0. \tag{8}$$

The plain prior charges for a uniform shift. But a uniform shift of all β_i is a change of B_g, a direction the tied model leaves completely free. If the intermediate model kept this penalty, it would constrain B_g and the comparison between the models would be unfair.

A shape change. Take $\Delta\beta = a\,(1,-1)^{\mathsf{T}}$: the bins move apart. The same algebra gives

$$\Delta\beta^{\mathsf{T}} \mathsf{C}^{-1} \Delta\beta \;=\; \frac{2a^2}{s^2\,(1-r)}. \tag{9}$$

This is the penalty the prior is *meant* to apply: it grows without bound as $r \to 1$, where the two bins are forced to track each other, and it relaxes to the independent value $2a^2/s^2$ as $r \to 0$.

A number makes the contrast concrete. At $r = 0.670$ (the worked value of Section 3), the shape change costs $2a^2/(0.330\,s^2) = 6.1\,a^2/s^2$ while the uniform shift costs $2a^2/(1.670\,s^2) = 1.2\,a^2/s^2$. The correlated prior already charges five times more for the shape change than for the uniform shift; the projection below takes the uniform-shift charge all the way to zero.

The projected precision. The repair for the uniform shift is to subtract its contribution from the inverse covariance. Define, with $\mathbf{1} = (1, 1)^\mathsf{T}$ the ones vector,

$$\mathsf{P} \;=\; \mathsf{C}^{-1} \;-\; \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}}{\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1}}. \tag{10}$$

Section 4 explains where this formula comes from; here, compute it. Because the two rows of C^{-1} have the same sum, $\mathsf{C}^{-1}\mathbf{1} = \big[s^2(1+r)\big]^{-1}(1, 1)^\mathsf{T}$ and $\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1} = 2/\big[s^2(1+r)\big]$. Substituting into Eq. (10) and simplifying each entry,

$$\mathsf{P} \;=\; \frac{1}{2s^2(1-r)}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \Delta\beta^\mathsf{T}\mathsf{P}\,\Delta\beta \;=\; \frac{(\beta_1 - \beta_2)^2}{2s^2(1-r)}. \tag{11}$$

The projected prior charges only for the *difference* of the two amplitudes. A uniform shift, $\beta_1 = \beta_2$, is now free, as in the tied model; a shape change keeps exactly the penalty of Eq. (9).

The denominator is not an arbitrary constant. With $\mathrm{Var}(\beta_i) = s^2$ and $\mathrm{Cov}(\beta_1, \beta_2) = s^2 r$,

$$\mathrm{Var}(\beta_1 - \beta_2) \;=\; s^2 + s^2 - 2s^2 r \;=\; 2s^2(1-r)\,: \tag{12}$$

the projected penalty is the squared jump between the bins divided by the prior variance of that jump. The whole construction reduces to one sentence: *the prior constrains the jump, not the level.*

3 Correlation, wandering, and the jump between neighbors

The kernel of Eq. (3) has two parameters, and they control two different things. The diagonal gives the wander of a single bin,

$$\mathrm{Var}(\delta_i) \;=\; \sigma_b^2, \tag{13}$$

so σ_b sets how far any one bin may drift from the group's passive-evolution trend. The jump between two bins separated by Δz follows as in Eq. (12):

$$\mathrm{Var}(\delta_i - \delta_j) \;=\; 2\,\sigma_b^2\left(1 - e^{-\Delta z/L}\right). \tag{14}$$

For separations much smaller than the correlation length, $\Delta z \ll L$, the exponential expands as $e^{-x} \approx 1 - x$, and

$$\mathrm{Var}(\delta_i - \delta_j) \;\approx\; \frac{2\,\sigma_b^2\,\Delta z}{L}, \qquad \text{rms jump} \;\approx\; \sigma_b\sqrt{\frac{2\,\Delta z}{L}}. \tag{15}$$

So L sets the correlation between bins, σ_b sets the wander of each bin, and *together* they set the jump a thin bin may take relative to its neighbor.

The combination that controls small separations is the roughness

$$\kappa \;\equiv\; \frac{\sigma_b}{\sqrt{L}}, \qquad \mathrm{Var}(\delta_i - \delta_j) \;\approx\; 2\,\kappa^2\,\Delta z. \tag{16}$$

The second form is the signature of diffusion: the variance of the increment grows linearly with the separation, exactly as the distance walked by a random walker grows linearly in time. On scales well below L the deviation curve $\delta(z)$ therefore looks like a random walk with diffusion coefficient κ^2; only on scales beyond L does the finite wander σ_b pull it back toward zero.

Worked example. Take neighboring thin bins with $\Delta z = 0.02$ and the baseline correlation length $L = 0.05$. Then $\Delta z/L = 0.4$ and

$$r \;=\; e^{-0.4} \;=\; 0.670. \tag{17}$$

The exact rms jump from Eq. (14) is $\sigma_b\sqrt{2\,(1-0.670)} = 0.81\,\sigma_b$: for $\sigma_b = 10\%$ the neighboring bins may differ by 8.1% (one sigma), and for $\sigma_b = 1\%$ by 0.81%. The Taylor form of Eq. (15) gives instead $\sigma_b\sqrt{0.8} \;=\; 0.89\,\sigma_b$, an 8.9% jump. The approximation overestimates because $\Delta z/L = 0.4$ is not actually small: the expansion replaces $1 - e^{-x}$ by x, and x exceeds $1 - e^{-x}$ already at second order ($x^2/2$), by about ten percent here.

4 Why the projection "sees" the uniform shift: three views

Equation (10) subtracts from C^{-1} a matrix built out of the single vector $v \;=\; \mathsf{C}^{-1}\mathbf{1}$: the subtracted piece is $v\,v^{\mathsf{T}}$ divided by the number $\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$. A matrix built from one vector acts along one direction only, so the subtraction removes the prior's information about exactly one direction, the uniform shift, and leaves everything else alone. This section derives the formula three times, each derivation adding its own intuition. Throughout, a group has n bins; $\Delta\beta$ and $\mathbf{1}$ are vectors with n entries.

4.1 View (a): subtracting the best-fitting uniform shift

Ask the following question: given a displacement $\Delta\beta$, how much of it can be explained as a pure change of the group amplitude? A change of B_g by c moves every β_i by c, so the candidate explanation is $c\,\mathbf{1}$, and the unexplained part should be judged by the prior metric. Minimize over the scalar c the quadratic form

$$\chi^2(c) \;=\; \big(\Delta\beta - c\,\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\Delta\beta - c\,\mathbf{1}\big). \tag{18}$$

Differentiating with respect to c (the two cross terms are equal because C^{-1} is symmetric),

$$\frac{d\chi^2}{dc} \;=\; -2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\big(\Delta\beta - c\,\mathbf{1}\big) \;=\; 0, \qquad \frac{d^2\chi^2}{dc^2} \;=\; 2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \;>\; 0, \tag{19}$$

where the second derivative is positive because C^{-1} is positive definite, so the solution of the first equation is the minimum of $\chi^2(c)$. That solution is

$$a \;=\; \frac{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}. \tag{20}$$

For a diagonal C with variances s_i^2 this reduces to $a = \sum_i \Delta\beta_i/s_i^2 \big/ \sum_i 1/s_i^2$, the familiar inverse-variance weighted mean; Eq. (20) is its generalization to correlated errors: generalized least squares for a constant.

Keep the dimensions straight: $\Delta\beta$, $\mathbf{1}$ and the residual $w = \Delta\beta - a\,\mathbf{1}$ are vectors with n entries; a is a single number. The residuals w satisfy one linear constraint (Eq. (22) below), so they span $n-1$ dimensions: one direction of the n-dimensional displacement space has been handed to the amplitude, and $n-1$ directions of genuine shape freedom remain.

4.2 View (b): a Pythagorean split of the penalty

Write the displacement as the best uniform shift plus a residual,

$$\Delta\beta \;=\; a\,\mathbf{1} + w. \tag{21}$$

The first-order condition of Eq. (19) says precisely that

$$\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}w \;=\; 0\,: \tag{22}$$

the residual is orthogonal to the uniform direction *in the* C^{-1} *metric*. (With correlated errors, "orthogonal" must be measured with the same matrix that measures lengths.) Substituting the split into the full penalty, the cross term is $2a\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}w = 0$, and

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; a^2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \;+\; w^{\mathsf{T}}\mathsf{C}^{-1}w. \tag{23}$$

This is the Pythagorean theorem in the prior metric: the squared length of the displacement is the squared length of its uniform piece plus the squared length of its shape piece.

Now compare with the projection. Inserting Eq. (20) into the subtracted term of Eq. (10) shows that the subtraction removes exactly $a^2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$, the uniform piece. What remains is

$$\Delta\beta^{\mathsf{T}}\mathsf{P}\,\Delta\beta \;=\; w^{\mathsf{T}}\mathsf{C}^{-1}w. \tag{24}$$

Two consequences follow immediately. First, a pure uniform shift has $w = 0$, so its penalty vanishes. Equivalently $\mathsf{P}\,\mathbf{1} = 0$: the projected matrix sends the uniform direction to zero and charges nothing for it. Second, a pure shape change (a displacement that already satisfies Eq. (22)) has $a = 0$ and keeps its original penalty untouched. The projection removes one direction and alters nothing else.

4.3 View (c): marginalizing a flat-prior amplitude

The third view explains why the projection, rather than some other uniform-shift repair, is the *correct* prior. Model the group amplitudes as

$$\beta \;=\; B\,\mathbf{1} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \mathsf{C}), \tag{25}$$

with a flat prior on the scalar B. This is literally the intermediate model of Eq. (2), linearized around the fiducial. The prior density of β is obtained by integrating B out:

$$p(\beta) \;\propto\; \int dB\, \exp\!\Big[-\tfrac{1}{2}\big(\beta - B\,\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\beta - B\,\mathbf{1}\big)\Big]. \tag{26}$$

The exponent is quadratic in B. Writing $S = \mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$ and completing the square,

$$\big(\beta - B\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\beta - B\mathbf{1}\big) \;=\; S\,\big(B - B_*\big)^2 \;+\; \beta^{\mathsf{T}}\mathsf{C}^{-1}\beta \;-\; \frac{\big(\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\beta\big)^2}{S}, \qquad B_* = \frac{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\beta}{S}. \tag{27}$$

The B integral is a Gaussian centered on B_* with width $1/\sqrt{S}$; it contributes only a β-independent constant $\sqrt{2\pi/S}$. What survives in the exponent is

$$p(\beta) \;\propto\; \exp\Big[-\tfrac{1}{2}\,\beta^{\mathsf{T}}\mathsf{P}\,\beta\Big], \qquad \mathsf{P} \;=\; \mathsf{C}^{-1} - \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}, \tag{28}$$

exactly the projection of Eq. (10).

The same computation has a standard matrix name. Expanding the exponent of Eq. (26) in the joint variables (β, B) gives a Gaussian with the $(n+1)\times(n+1)$ block precision

$$\mathsf{F}_{(\beta,B)} \;=\; \begin{pmatrix} \mathsf{C}^{-1} & -\,\mathsf{C}^{-1}\mathbf{1} \\ -\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1} & \mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \end{pmatrix}. \tag{29}$$

Removing a parameter from a precision matrix is not just deleting its row and column: the remaining block must also be corrected downward by whatever the removed parameter could absorb. The corrected block, $\mathsf{P} = \mathsf{F}_{\beta\beta} - \mathsf{F}_{\beta B}\,\mathsf{F}_{BB}^{-1}\,\mathsf{F}_{B\beta}$, is called the *Schur complement* over B, and it reproduces Eq. (10) term by term. So P is not a trick: it is precisely the statement "B_g *is sampled with a flat prior and marginalized*", folded into one matrix.

Note also that B_* in Eq. (27) is the same generalized-least-squares mean as a in Eq. (20): the marginalization integrates over all amplitudes, and the best-fitting one dominates the integral.

4.4 Checking views (b) and (c) on the two-bin case

For $n = 2$ the rows of C^{-1} in Eq. (6) have equal sums, so Eq. (20) gives the plain mean, $a = (\Delta\beta_1 + \Delta\beta_2)/2$, and the residual is the antisymmetric half-difference, $w = \frac{1}{2}(\Delta\beta_1 - \Delta\beta_2)\,(1,-1)^{\mathsf{T}}$. Its penalty, using the shape-change result of Eq. (9) with $a \to \frac{1}{2}(\Delta\beta_1 - \Delta\beta_2)$, is

$$w^{\mathsf{T}}\mathsf{C}^{-1}w \;=\; \frac{2}{s^2(1-r)}\left(\frac{\Delta\beta_1 - \Delta\beta_2}{2}\right)^2 \;=\; \frac{(\Delta\beta_1 - \Delta\beta_2)^2}{2s^2(1-r)}, \tag{30}$$

which is the direct two-bin computation of Eq. (11). Views (a), (b) and (c) all land on the same matrix.

5 Generalization to many bins and several groups

5.1 Three bins: total correlation versus direct links

A group now has n bins at redshifts $z_1 < z_2 < \cdots < z_n$, unevenly spaced in general, with the prior covariance of Eq. (5),

$$\mathsf{C}_{ij} \;=\; s^2 \exp\!\left(-\,\frac{|z_i - z_j|}{L}\right). \tag{31}$$

This matrix is positive definite for any set of distinct redshifts (all its eigenvalues are positive), so the prior is a proper probability distribution and C^{-1} exists.

The structure of C^{-1} is best seen on three evenly spaced bins. With neighbor correlation $r = e^{-\Delta z/L}$, the outer pair sits two steps apart and carries $e^{-2\Delta z/L} = r^2$, so

$$\mathsf{C} \;=\; s^2\begin{pmatrix} 1 & r & r^2 \\ r & 1 & r \\ r^2 & r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)}\begin{pmatrix} 1 & -r & 0 \\ -r & 1+r^2 & -r \\ 0 & -r & 1 \end{pmatrix}. \tag{32}$$

One multiplication confirms the surprising entry. The $(1,3)$ entry of $\mathsf{C}^{-1}\mathsf{C}$ must be zero (it is an off-diagonal entry of the identity matrix), and indeed row 1 of C^{-1} times column 3 of C gives, dropping the prefactors,

$$1 \cdot r^2 \;+\; (-r) \cdot r \;+\; 0 \cdot 1 \;=\; r^2 - r^2 \;=\; 0. \tag{33}$$

Here is the point that looks paradoxical at first. Bins 1 and 3 *are* correlated (C contains r^2), yet their entry in C^{-1} is exactly zero. The two matrices answer different questions. C gives the *total* correlation: what δ_1 tells you about δ_3 when nothing else is known. C^{-1} gives the *direct* correlation: what δ_1 still tells you about δ_3 once δ_2 is already known, and that is nothing. All of the 1–3 correlation travels through bin 2.

Think of a chain of people passing a message. The first person's message reaches far down the chain, but the links are only between neighbors: nobody talks past the person standing next to them.

The exponential kernel is special in making this exact, because its correlations multiply along the chain:

$$e^{-(z_3-z_1)/L} \;=\; e^{-(z_3-z_2)/L}\, e^{-(z_2-z_1)/L}. \tag{34}$$

The long-distance correlation is the product of the step correlations, so the middle bin already carries everything the outer bins share. A Gaussian kernel $e^{-\Delta z^2/L^2}$ does not factorize this way, because the squared separations do not add up along the chain: $(z_3-z_1)^2 \neq (z_3-z_2)^2+(z_2-z_1)^2$. Its inverse is dense.

The same structure shows directly in the penalty. For n evenly spaced bins, the quadratic form of the deviations δ_i (whose covariance is $\sigma_b^2\mathsf{R}$) regroups into a sum over neighbor links,

$$\delta^{\mathsf{T}}\big(\sigma_b^2\mathsf{R}\big)^{-1}\delta \;=\; \frac{\delta_1^2}{\sigma_b^2} \;+\; \sum_{k=1}^{n-1} \frac{\big(\delta_{k+1} - r\,\delta_k\big)^2}{\sigma_b^2\,(1-r^2)}. \tag{35}$$

Each term compares one bin with r times the bin before it: every bin is pulled only by its neighbor, and the long-range correlation comes from chaining the pulls. (Expanding the squares and collecting terms reproduces Eq. (32) for $n = 3$; Appendix A does it for general, uneven spacing.) This is the precise sense of "correlation only between adjacent bins": *directly*, only neighbors talk; *in total*, the influence propagates down the whole chain. Probability theory calls this neighbors-only structure the Markov property.

5.2 The tridiagonal inverse for n bins

A matrix whose nonzero entries sit only on the diagonal and the two neighboring off-diagonals is called *tridiagonal*. The three-bin inverse of Eq. (32) is the smallest interesting example, and the pattern grows unchanged. For n evenly spaced bins, where every adjacent pair has the same correlation $r = e^{-\Delta z/L}$ and $\mathsf{C}_{ij} = s^2\, r^{|i-j|}$,

$$\mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)} \begin{pmatrix} 1 & -r & & & \\ -r & 1+r^2 & -r & & \\ & -r & \ddots & \ddots & \\ & & \ddots & 1+r^2 & -r \\ & & & -r & 1 \end{pmatrix}, \tag{36}$$

with diagonal $(1,\ 1+r^2,\ \ldots,\ 1+r^2,\ 1)$, off-diagonal entries $-r$, blanks zero, and the overall factor $1/\big[s^2(1-r^2)\big]$. The entries read off the chain sum of Eq. (35). An off-diagonal entry is the link between that pair of neighbors. An interior bin touches two links, one to the bin on its left and one to the bin on its right, and the two pulls together contribute $1+r^2$; the first and last bins touch only one link each, hence the plain 1 in the corners.

With uneven spacing, each adjacent pair has its own step and therefore its own correlation,

$$r_k \;=\; \exp\!\left(-\,\frac{z_{k+1}-z_k}{L}\right), \qquad k=1,\ldots,n-1, \tag{37}$$

and the inverse keeps the same shape with per-pair entries:

$$s^2\left(\mathsf{C}^{-1}\right)_{kk} = \begin{cases} \dfrac{1}{1-r_1^2}, & k=1, \\ \dfrac{1}{1-r_{k-1}^2}+\dfrac{r_k^2}{1-r_k^2}, & 1<k<n, \\ \dfrac{1}{1-r_{n-1}^2}, & k=n, \end{cases} \qquad s^2\left(\mathsf{C}^{-1}\right)_{k,k+1} = \frac{-\,r_k}{1-r_k^2}. \tag{38}$$

In words: the off-diagonal entry is the neighbor link of pair k, stronger when the bins are closer (r_k nearer one); the diagonal entry of bin k collects the links touching that bin, one term from the pair on its left and one from the pair on its right; the first and last bins carry one link only. Setting all $r_k = r$ turns the interior diagonal into $1/(1-r^2)+r^2/(1-r^2) = (1+r^2)/(1-r^2)$ and recovers Eq. (36). Appendix A derives Eq. (38) from the chain of neighbor links; the companion script `verify_prior.py` checks the three-bin, even and uneven forms numerically.

5.3 The projection, group by group

Each group receives the projected precision of Eq. (10), built from its own C. The properties derived in Section 4 carry over verbatim to n bins. $\mathsf{P}\,\mathbf{1} = 0$: a uniform shift of the whole group costs nothing. P is positive semidefinite, which means no displacement is *rewarded*: every move either costs χ^2 or is free. And of the n directions a group can move in, exactly one (the uniform shift) is free while the other $n-1$ are charged: in matrix language, P has rank $n-1$.

Several groups combine trivially. The prior of Eq. (3) is zero across groups, so the full prior covariance is block-diagonal, one block per group; and each group has its own free amplitude B_g, so each block is projected separately. The total prior precision is the block-diagonal matrix of the per-group projections.

5.4 The two limits

The intermediate model contains both extremes.

As $L \to \infty$, every $r_k \to 1$, so the jump variance $2\sigma_b^2(1-r_k)$ of every adjacent pair goes to zero: the prior forces all deviations of a group to be equal. The only freedom left is their common value, a uniform shift, and that is the direction the projection leaves free. This is the tied model, with one free B_g per group, *whatever* the value of σ_b.

As $\sigma_b \to \infty$ at fixed L, the overall scale of C diverges, so $\mathsf{C}^{-1} \to 0$ and with it $\mathsf{P} \to 0$: the prior penalty vanishes and every bin's amplitude is free: the independent model with flat priors.

Between the limits, L and σ_b interpolate continuously, which is what makes the model useful: the figure of merit can be traced as a function of how much bin-to-bin freedom the bias is granted.

6 How the prior enters the Fisher matrix

A Gaussian prior adds its precision to the Fisher matrix:

$$F \;=\; F_{\rm data} \;+\; F_{\rm priors}, \tag{39}$$

where $F_{\rm data}$ is the usual data Fisher matrix and $F_{\rm priors}$ holds one block per prior. The projected precision P of each group is added to the rows and columns of that group's bin amplitudes β_i. All other parameters (the cosmological parameters, and the velocity dispersions with their own independent Gaussian priors) are unchanged: P has no entries outside the bias block.

One consequence matters in practice. Bins that lost all their band powers to the scale cut carry no data, so their rows of $F_{\rm data}$ are zero. In the independent model such a parameter is completely unconstrained and must be dropped from the matrix before inversion. With the correlated prior it stays: P ties the empty bin to its neighbors, which do carry data, so the matrix remains invertible and the empty bin follows the chain.

Choice of the hyperparameters. The forecast scans

$$L \in \{0.02,\ 0.05,\ 0.1,\ 0.2\} \quad \text{(in redshift)}, \qquad \sigma_b \in \{1\%,\ 5\%,\ 10\%,\ 20\%\}, \tag{40}$$

with baseline $L = 0.05$ and $\sigma_b = 10\%$, and the plausible region $L = 0.02$–0.1, $\sigma_b = 5$–20%. The upper end of L is physical, not numerical: a sample's flux limit and color cuts change which galaxies enter the sample as redshift grows, and that change of galaxy mix happens over $\Delta z \sim 0.1$. Bias deviations should therefore not be assumed coherent over separations much beyond that, so $L \lesssim 0.1$.

Numerical verification. The companion script `verify_prior.py`, in this folder, checks every identity used above: the two-bin closed forms of Eqs. (6) and (11); the three-bin inverse of Eq. (32), including its exactly zero $(1,3)$ entry; the null direction, positive semidefiniteness and rank $n-1$ of P on uneven redshifts; the tridiagonal inverses of Eqs. (36) and (38); the Pythagorean identity of Eq. (24) on random displacements; the equality of P with the Schur complement of the joint (β, B) precision of Eq. (29); the jump variance of Eq. (14) by Monte Carlo (10^5 draws); and the worked example of Section 3. Each check prints a PASS or FAIL line with its measured deviation.

Summary

The intermediate bias model frees one amplitude B_g per tracer group and lets each bin deviate fractionally from the group's passive-evolution trend $B_g/D(z)$, with deviations correlated over a redshift length L and a per-bin wander σ_b. In the Fisher matrix this becomes one projected precision P per group, added to the bin-amplitude block: P charges for differences between neighboring amplitudes but never for a uniform shift of the group, links only neighboring bins directly (the inverse covariance is tridiagonal, even though the correlation itself reaches far), and reduces to the tied model as $L \to \infty$ and to independent flat-prior biases as $\sigma_b \to \infty$.

A The tridiagonal inverse from the chain of links

Because correlations multiply along the chain (Eq. (34)), the prior can be built one bin at a time: the joint density of the deviations factorizes into neighbor conditionals,

$$p(\delta_1, \ldots, \delta_n) \;=\; p(\delta_1)\, p(\delta_2 \mid \delta_1) \cdots p(\delta_n \mid \delta_{n-1}), \tag{41}$$

where each conditional is Gaussian with

$$\big\langle \delta_{k+1} \mid \delta_k \big\rangle \;=\; r_k\, \delta_k, \qquad \mathrm{Var}\big(\delta_{k+1} \mid \delta_k\big) \;=\; \sigma_b^2\, \big(1 - r_k^2\big). \tag{42}$$

Here $r_k = e^{-(z_{k+1}-z_k)/L}$ is, as in Sections 3 and 5.2, the correlation coefficient of the adjacent bins k and $k{+}1$: the kernel of Eq. (3) gives $\mathrm{Cov}(\delta_k, \delta_{k+1}) = \sigma_b^2\, r_k$ and $\mathrm{Var}(\delta_k) = \mathrm{Var}(\delta_{k+1}) = \sigma_b^2$.

The conditional mean and variance in Eq. (42) follow from nothing more than these covariances. Split δ_{k+1} into the part predicted by its neighbor and a leftover e,

$$\delta_{k+1} \;=\; a\, \delta_k + e, \tag{43}$$

and choose the number a so that the leftover carries no correlation with δ_k:

$$\mathrm{Cov}(e, \delta_k) \;=\; \mathrm{Cov}(\delta_{k+1}, \delta_k) - a\, \mathrm{Var}(\delta_k) \;=\; \sigma_b^2\, r_k - a\, \sigma_b^2 \;=\; 0 \quad\Longrightarrow\quad a = r_k. \tag{44}$$

The leftover's variance follows by expanding $e = \delta_{k+1} - r_k \delta_k$:

$$\mathrm{Var}(e) \;=\; \mathrm{Var}(\delta_{k+1}) - 2 r_k\, \mathrm{Cov}(\delta_{k+1}, \delta_k) + r_k^2\, \mathrm{Var}(\delta_k) \;=\; \sigma_b^2 - 2 r_k^2 \sigma_b^2 + r_k^2 \sigma_b^2 \;=\; \sigma_b^2\, \big(1 - r_k^2\big). \tag{45}$$

For jointly Gaussian variables, uncorrelated means independent. So once δ_k is known, the only randomness left in $\delta_{k+1} = r_k \delta_k + e$ is e itself: the conditional mean is $r_k \delta_k$ and the conditional variance is $\sigma_b^2(1 - r_k^2)$, which is Eq. (42). In one line: knowing the neighbor explains the fraction r_k^2 of the bin's variance, and the fraction $1 - r_k^2$ is new.

The factorization of Eq. (41) needs one more fact: conditioning on δ_k alone must be enough: the earlier bins must add nothing. This is again the multiplication of correlations along the chain. For any earlier bin $j \le k$,

$$\mathrm{Cov}(e, \delta_j) \;=\; \sigma_b^2 \left[e^{-(z_{k+1}-z_j)/L} - r_k\, e^{-(z_k - z_j)/L} \right] \;=\; \sigma_b^2\, e^{-(z_k - z_j)/L} \left[e^{-(z_{k+1}-z_k)/L} - r_k \right] \;=\; 0, \tag{46}$$

because the first exponential factorizes through z_k and the bracket is $r_k - r_k$. The leftover is therefore independent of every earlier bin, and the chain of one-step conditionals is the complete prior.

Taking $-2 \ln p$ of Eq. (41) gives the penalty as the chain sum, the uneven-spacing version of Eq. (35):

$$\delta^{\mathsf{T}} \big(\sigma_b^2 \mathsf{R}\big)^{-1} \delta \;=\; \frac{\delta_1^2}{\sigma_b^2} \;+\; \sum_{k=1}^{n-1} \frac{\big(\delta_{k+1} - r_k\, \delta_k\big)^2}{\sigma_b^2\, \big(1 - r_k^2\big)}. \tag{47}$$

Expanding each square and collecting the coefficient of every product $\delta_i \delta_j$ reads off the precision matrix entry by entry. δ_1^2 appears twice: in the first term (coefficient 1) and in the $k = 1$ link as the predictor (coefficient $r_1^2/(1{-}r_1^2)$); the two add up to $1/(1{-}r_1^2)$. An interior δ_k^2 appears in the $k{-}1$ link as the predicted bin ($1/(1{-}r_{k-1}^2)$) and in the k link as the predictor ($r_k^2/(1{-}r_k^2)$). The last bin is only ever predicted: $1/(1 - r_{n-1}^2)$. Each product $\delta_k \delta_{k+1}$ appears in its own link only, with coefficient $-2r_k/(1 - r_k^2)$, i.e. a symmetric matrix entry $-r_k/(1 - r_k^2)$. No product of non-neighbors ever appears, so the matrix is tridiagonal by construction. This is Eq. (38) in units of σ_b^2; replacing δ by $\Delta\beta = B_g\, \delta$ and σ_b by $s = \sigma_b B_g$ gives the same matrix for the bin amplitudes.